Medvale Internal Medicine

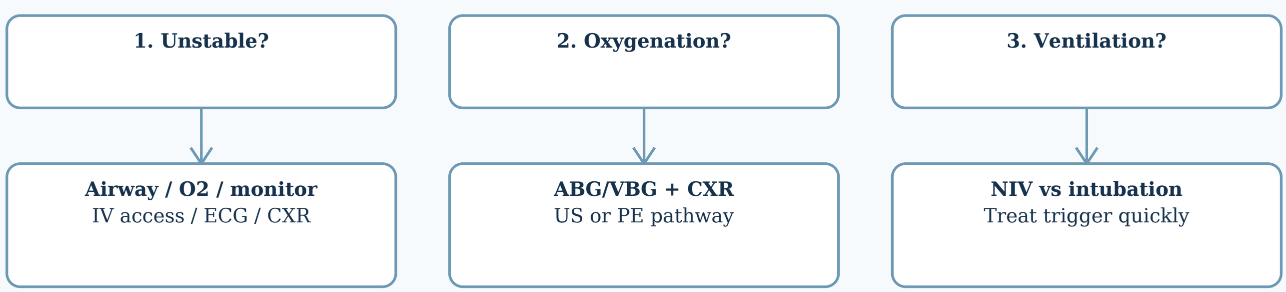
Chapter 2 - Pulmonary Medicine: Approach to Dyspnea, Hypoxemia, and Respiratory Failure

Purpose. This chapter is an original Medvale teaching chapter. The uploaded Anki CSV was used only as a coverage signal: it showed recurring pulmonary concepts such as asthma/COPD physiology, pneumonia, pleural disease, pneumothorax, pulmonary embolism, lung cancer, pulmonary function testing, hypoxemia, hypercapnia, and ARDS. The wording, tables, diagrams, clinical framing, and practice questions below are newly written for Medvale.

Why this matters clinically

Dyspnea is one of the highest-yield internal medicine presentations because it is not a diagnosis; it is a physiologic alarm. The same complaint can represent asthma, acute coronary syndrome, pulmonary embolism, pneumonia, metabolic acidosis, anemia, panic, neuromuscular weakness, or early sepsis. A strong pulmonary chapter therefore begins with a reasoning framework that separates oxygenation failure from ventilation failure, parenchymal disease from airway disease, and pulmonary disease from cardiac or metabolic mimics. In the emergency department and inpatient setting, this distinction determines whether the first intervention is oxygen, bronchodilation, antibiotics, diuresis, anticoagulation, noninvasive ventilation, intubation, or drainage of pleural air or fluid.

A second reason this framework matters is that respiratory compensation can hide dangerous disease. A young patient with pneumonia may maintain a normal oxygen saturation while breathing 34 times per minute. A patient with COPD may look comfortable despite a high carbon dioxide level because chronic compensation has raised serum bicarbonate. A patient with pulmonary embolism may have a clear chest radiograph but profound tachycardia and pleuritic pain. For exams and clinical practice, the goal is to read the pattern rather than memorize isolated facts.



Work in parallel: stabilize physiology while the history, exam, ECG, CXR, and first labs localize the syndrome.

Core illness scripts

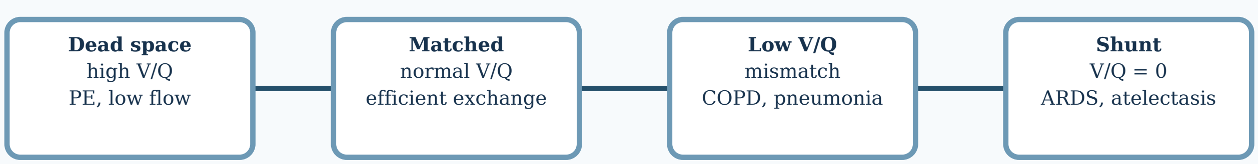
Syndrome	Classic pattern	First bedside question	Immediate priority
Upper airway obstruction	Stridor, voice change, drooling, facial/neck swelling, abrupt distress	Can air move through the upper airway?	Call for airway help; prepare advanced airway rather than delaying for imaging.
Asthma/COPD exacerbation	Wheeze, prolonged expiration, accessory muscle use; COPD often has chronic sputum or smoking history	Is this bronchospasm with rising work of breathing?	Short-acting bronchodilator therapy, systemic steroids when appropriate, oxygen titrated to target, NIV if ventilatory failure.

Syndrome	Classic pattern	First bedside question	Immediate priority
Pneumonia	Fever, cough, sputum, pleuritic pain, focal crackles, infiltrate	Is there infection plus alveolar filling?	Assess severity, obtain cultures when indicated, start empiric antibiotics promptly in ill patients.
Pulmonary embolism	Pleuritic pain, tachycardia, dyspnea, hemoptysis/syncope; risk factors for thrombosis	Is pretest probability high enough to image or treat?	Risk-stratify; anticoagulate when appropriate; consider thrombolysis/embolectomy for obstructive shock.
Pulmonary edema	Orthopnea, crackles, S3, hypertension or ischemia, bilateral opacities	Is fluid flooding alveoli from high hydrostatic pressure?	Oxygen/NIV, diuresis and afterload reduction when cardiogenic, treat trigger.
Pneumothorax	Sudden unilateral pleuritic pain and dyspnea; decreased breath sounds; tension physiology if hypotensive	Is trapped pleural air impairing ventilation or venous return?	Needle/chest decompression for tension; chest tube or observation depending size and stability.
ARDS	Acute severe hypoxemia after sepsis, aspiration, trauma, pancreatitis, transfusion; diffuse bilateral opacities	Is this inflammatory noncardiogenic alveolar flooding?	Lung-protective ventilation, treat cause, conservative fluids when stable.
Neuromuscular ventilatory failure	Weak cough, bulbar symptoms, shallow breathing, paradoxical abdomen, normal lungs early	Is the respiratory pump failing?	Measure vital capacity/NIF, early ventilatory support, treat neurologic cause.

Pathophysiology: the four questions of gas exchange

The lungs have two jobs: move oxygen into blood and remove carbon dioxide from blood. Oxygenation depends heavily on matching alveolar ventilation to pulmonary perfusion, preserving an intact alveolar-capillary surface, maintaining adequate inspired oxygen, and avoiding true shunt. Carbon dioxide removal depends more directly on alveolar ventilation. Because carbon dioxide diffuses easily, a high $PaCO_2$ usually means the patient is not ventilating enough for metabolic demand, not merely that oxygen transfer is impaired.

A practical way to analyze dyspnea is to ask four questions. First, is there enough inspired oxygen reaching the alveoli? This fails at altitude, with low FiO_2, or when severe airway obstruction prevents delivery. Second, are ventilated alveoli perfused? Pulmonary embolism creates relatively high V/Q or dead space physiology. Third, are perfused alveoli ventilated? Pneumonia, edema, atelectasis, and COPD create low V/Q units. Fourth, is blood bypassing ventilated alveoli entirely? A large shunt, such as lobar atelectasis or severe ARDS, responds poorly to oxygen until lung units are recruited or the underlying process reverses.



Oxygen improves most V/Q mismatch; a large true shunt improves poorly until collapsed or fluid-filled units are reopened or drained.

Mechanisms of hypoxemia

Mechanism	A-a gradient	Response to oxygen	Examples	Exam clue
Low inspired oxygen	Normal	Improves	High altitude, low FiO2 delivery error	Multiple patients or setting clue; lungs may be normal.
Hypoventilation	Normal	Improves if ventilation improves	Opioids, CNS depression, neuromuscular weakness, severe obesity hypoventilation	High PaCO2; oxygen alone may not fix ventilatory failure.
V/Q mismatch	High	Usually improves	COPD/asthma, pneumonia, pulmonary edema, PE	Most common pattern; CXR, exam, and risk factors guide cause.
Diffusion limitation	High, worse with exertion	Improves	Interstitial lung disease, emphysema with low DLCO	Exertional desaturation; resting values may be deceptively mild.
Right-to-left shunt	High	Poor or incomplete response	ARDS, atelectasis, intracardiac shunt, AVM	Severe hypoxemia despite high-flow oxygen suggests shunt.

ABG and VBG interpretation at the bedside

Arterial blood gas analysis is most valuable when the clinical question is oxygenation, acid-base status, or severity of ventilatory failure. Venous blood gas can approximate pH and PaCO2 trends in many settings, but it cannot replace arterial PaO2 when oxygenation severity is central. A useful first pass is: look at pH, then PaCO2, then bicarbonate, then PaO2 and the clinical oxygen requirement. Respiratory acidosis means carbon dioxide retention. Respiratory alkalosis means excessive ventilation relative to carbon dioxide production, often due to pain, anxiety, early sepsis, hypoxemia, pulmonary embolism, pregnancy, or liver disease.

Pattern	Typical interpretation	Common causes	Management implication
Low pH + high PaCO2	Acute or acute-on-chronic ventilatory failure	COPD/asthma exhaustion, opioid toxicity, neuromuscular weakness, severe obesity hypoventilation	Support ventilation: NIV when appropriate; intubate if airway protection, shock, severe acidosis, or fatigue.
High pH + low PaCO2	Respiratory alkalosis	PE, early pneumonia/sepsis, pain, anxiety, pregnancy, cirrhosis	Do not dismiss as anxiety until dangerous triggers are excluded.
Low PaO2 + normal/low PaCO2	Primary oxygenation failure	Pneumonia, edema, PE, ARDS	Escalate oxygen delivery; search for parenchymal, vascular, or shunt cause.
Low PaO2 + high PaCO2	Combined oxygenation and ventilation failure	Severe COPD/asthma, advanced neuromuscular disease, severe chest wall disease	Treat both gas exchange and respiratory pump failure.
Normal SpO2 but marked tachypnea	Possible compensation before decompensation	Sepsis, metabolic acidosis, early PE, early pneumonia	Check work of breathing and metabolic data; normal saturation is not reassurance alone.

Diagnostic approach

The initial diagnostic approach should be simultaneous rather than sequential. While oxygen and monitoring are started, the clinician should obtain vital signs, focused history, cardiopulmonary examination, pulse oximetry, ECG, chest radiograph, and basic laboratory studies tailored to severity. A patient with dyspnea and hypotension is a shock patient until proven otherwise; the pulmonary differential narrows quickly to tension pneumothorax, massive PE, severe asthma/COPD with dynamic hyperinflation, severe pneumonia/sepsis, tamponade, myocardial infarction, or decompensated heart failure.

The history should identify tempo. Seconds to minutes suggests airway obstruction, pneumothorax, PE, arrhythmia, aspiration, or panic. Hours to days suggests pneumonia, asthma/COPD exacerbation, pulmonary edema, or evolving PE. Weeks to months suggests malignancy, interstitial lung disease, anemia, chronic heart failure, pulmonary hypertension, occupational lung disease, or medication toxicity. Associated symptoms then localize the syndrome: fever and sputum point toward infection; orthopnea and edema toward heart failure; pleuritic pain and hemoptysis toward PE or pneumothorax; wheeze toward obstructive disease, though wheeze can also occur in pulmonary edema and foreign body obstruction.

First-pass diagnostic table

Finding	Most likely bucket	Why it matters
Stridor, drooling, muffled voice	Upper airway obstruction	This is an airway problem first; imaging should not delay airway planning.
Diffuse wheeze with prolonged expiration	Obstructive lower airway disease	Air trapping increases work of breathing and can progress to hypercapnia.
Crackles + orthopnea + edema	Pulmonary edema/heart failure	NIV and diuresis can rapidly improve oxygenation when cardiogenic.
Fever + focal findings + infiltrate	Pneumonia	Alveolar filling produces V/Q mismatch and can progress to sepsis or ARDS.
Clear lungs + pleuritic pain + tachycardia	Pulmonary embolism	CXR can be normal; risk-stratification drives D-dimer vs CT angiography.
Unilateral absent breath sounds + hypotension	Tension pneumothorax	Clinical decompression precedes confirmatory imaging in unstable patients.
Somnolence + bradypnea + miosis	Drug-induced hypoventilation	Ventilation and reversal therapy may be more important than oxygen alone.
Weak cough + normal lung exam + rising CO2	Neuromuscular failure	Early intubation may be safer than waiting for hypoxemia.

Pulmonary function tests as a reasoning tool

Pulmonary function testing is often taught as a set of numbers, but clinically it answers three questions. Is airflow leaving the lung too slowly? Is total lung capacity reduced? Is gas transfer across the alveolar-capillary membrane impaired? Obstruction is primarily an expiratory flow problem, so FEV1 falls disproportionately to FVC and the FEV1/FVC ratio decreases. Restriction is a volume problem, so total lung capacity is reduced. DLCO adds a gas-transfer dimension: it is low when alveolar surface area or pulmonary capillary blood volume is reduced, and it may be normal in pure extrapulmonary restriction such as obesity or neuromuscular weakness.

PFT pattern	FEV1/FVC	TLC	DLCO	Typical causes	Clinical interpretation
Obstructive	Low	Normal or high	Normal in asthma; low in emphysema	Asthma, COPD, bronchiectasis	Airflow limitation; bronchodilator response supports asthma but does not exclude COPD overlap.

PFT pattern	FEV1/FVC	TLC	DLCO	Typical causes	Clinical interpretation
Restrictive - parenchymal	Normal/high	Low	Low	Idiopathic pulmonary fibrosis, pneumoconiosis, sarcoidosis	Small stiff lungs with impaired gas transfer.
Restrictive - extrapulmonary	Normal/high	Low	Normal or mildly low	Obesity, kyphoscoliosis, neuromuscular weakness	Mechanical limitation; gas membrane may be intact.
Isolated low DLCO	Often normal	Often normal	Low	Early ILD, pulmonary vascular disease, emphysema, anemia	Think membrane, capillary bed, or hemoglobin problem.
Upper airway obstruction	Variable; flow-volume loop abnormal	Often normal	Usually normal	Vocal cord dysfunction, tracheal stenosis	Dyspnea/wheeze mimic; spirometry loop is the clue.

Key number anchors. A post-bronchodilator FEV1/FVC below about 0.70 is a common practical threshold for obstruction in COPD-oriented frameworks, while asthma is supported by variable symptoms and reversible airflow limitation such as an FEV1 increase of at least 12% and at least 200 mL after bronchodilator. These thresholds are anchors, not substitutes for clinical context.

Initial management: stabilize first, then localize

Management begins with the patient in front of you rather than the final diagnosis. The first tasks are to assess mental status, airway protection, work of breathing, oxygen saturation, perfusion, and trajectory. A patient who is tiring, confused, hypotensive, cyanotic, or unable to speak full sentences needs escalation before a complete workup. Oxygen can treat hypoxemia, but it does not ventilate. Noninvasive ventilation can unload respiratory muscles and improve ventilation in selected patients, especially COPD exacerbation and cardiogenic pulmonary edema. Intubation is favored when the airway is unprotected, secretions are uncontrolled, shock is present, severe acidosis is worsening, or the patient cannot cooperate with a mask.

Therapy	Best fit	Avoid or use caution	Clinical endpoint
Nasal cannula/simple mask	Mild hypoxemia with stable work of breathing	Do not rely on it for ventilatory failure	SpO2 target reached with low work of breathing.
High-flow nasal cannula	Hypoxemic respiratory failure needing high FiO2 or flow support	Delayed intubation in a deteriorating patient	Lower respiratory rate/work, improved oxygenation.
Noninvasive ventilation	COPD hypercapnia, cardiogenic pulmonary edema, selected obesity hypoventilation	Vomiting, inability to protect airway, facial trauma, severe encephalopathy, shock	Improving pH/PaCO2, comfort, mental status.
Invasive mechanical ventilation	Airway failure, refractory hypoxemia, severe fatigue, shock, unsafe NIV	Dynamic hyperinflation in asthma/COPD requires careful settings	Gas exchange and work of breathing stabilized while cause is treated.
Disease-specific therapy	Antibiotics, bronchodilators/steroids, anticoagulation, diuresis, drainage	Premature closure on one diagnosis	Physiology improves and underlying driver is controlled.

Definitive management by physiologic bucket

Bucket	Definitive actions	Do not miss
Bronchospasm/air trapping	Short-acting beta agonist plus anticholinergic therapy for severe exacerbations, systemic corticosteroids when indicated, magnesium for severe asthma in selected cases, controlled oxygen, NIV for COPD ventilatory failure.	Silent chest, rising $PaCO_2$, altered mental status, or exhaustion signals impending failure.
Alveolar filling from infection	Empiric antibiotics based on site and severity, source control if empyema, fluids/vasopressors for sepsis, oxygen escalation as needed.	Normal early CXR can occur in dehydration or early infection; reassess if clinical picture persists.
Hydrostatic pulmonary edema	NIV for distress, loop diuresis, nitrates when hypertensive and appropriate, ischemia/arrhythmia treatment.	Wheeze can be cardiac asthma; do not call every wheeze COPD.
Pulmonary vascular obstruction	Risk-stratify PE, anticoagulate unless contraindicated, consider reperfusion for high-risk PE with shock.	Clear lungs and a normal CXR do not rule out PE.
Pleural space disease	Thoracentesis for new significant effusion, chest tube for complicated parapneumonic effusion/empyema or clinically significant pneumothorax.	Tension pneumothorax is a clinical diagnosis in an unstable patient.
Respiratory pump failure	Measure vital capacity and inspiratory force, avoid excessive sedation, treat neuromuscular cause, support ventilation early.	Oxygen saturation can stay normal until late; CO_2 and mechanics reveal danger.

Complications and deterioration patterns

Pulmonary patients deteriorate through a limited number of pathways. Hypoxemic failure worsens when more lung units become flooded, collapsed, or poorly perfused. Hypercapnic failure worsens when respiratory muscles fatigue or minute ventilation falls. Obstructive diseases can deteriorate because of dynamic hyperinflation: the patient cannot exhale fully before the next breath, intrathoracic pressure rises, venous return falls, and hypotension may occur. Pneumonia can deteriorate through sepsis, empyema, lung abscess, or ARDS. Pulmonary embolism can deteriorate through right ventricular failure and obstructive shock. Pleural air under pressure can deteriorate through mediastinal shift and impaired venous return.

Early warning sign	Likely meaning	Action
Respiratory rate rising despite oxygen	Work of breathing is not controlled	Reassess diagnosis and escalate support.
Patient becomes quiet or less wheezy during severe asthma	Air movement may be falling	Prepare for critical care/intubation rather than celebrating.
Increasing somnolence in COPD or opioid exposure	CO_2 narcosis or global fatigue	Check gas, ventilate, reverse/treat cause.
Hypotension after intubation in obstructive disease	Dynamic hyperinflation or reduced venous return	Disconnect briefly if severe, adjust expiratory time, fluids/pressors as needed.
Persistent fever with pleural effusion	Complicated parapneumonic effusion/empyema	Diagnostic thoracentesis and drainage if indicated.
Hypoxemia poorly responsive to high FiO_2	Large shunt physiology	Recruit/drain/reverse underlying alveolar collapse or flooding.

Differentials and mimics

The best pulmonary clinicians constantly ask whether the lungs are the victim rather than the source. Metabolic acidosis can cause dramatic tachypnea with clear lungs because the respiratory system is compensating. Anemia can cause dyspnea with normal oxygen saturation because saturation measures hemoglobin saturation, not oxygen content. Acute coronary syndrome can present with dyspnea rather than chest pain, especially in older adults and patients with diabetes. Panic attacks can cause dyspnea and paresthesias, but panic should be diagnosed after dangerous cardiopulmonary causes have been considered. Vocal cord dysfunction can mimic asthma but often has inspiratory symptoms, throat tightness, and a flow-volume loop pattern suggesting upper airway obstruction.

High-yield exam clues

- Obstructive disease lowers the FEV1/FVC ratio; restriction requires a low total lung capacity, not just a low FVC on spirometry.
- Asthma is supported by variable symptoms and bronchodilator reversibility; COPD is supported by persistent obstruction in a patient with exposure history.
- Very high A-a gradient points toward V/Q mismatch, diffusion limitation, or shunt; a normal A-a gradient with hypoxemia suggests hypoventilation or low inspired oxygen.
- Pulmonary embolism often has dyspnea, pleuritic pain, tachycardia, and a relatively unrevealing lung exam or chest radiograph.
- Transudative pleural effusions reflect systemic pressure/oncotic problems; exudative effusions reflect local pleural inflammation, infection, malignancy, or embolic disease.
- Tension pneumothorax is treated before imaging if the patient is unstable.
- In neuromuscular respiratory failure, oxygen saturation may remain normal until late; declining vital capacity, weak cough, bulbar symptoms, and CO_2 retention are more important.
- ARDS is not simply any hypoxemia; it is acute inflammatory lung injury with bilateral opacities not fully explained by cardiac failure or fluid overload.

Common pitfalls

Pitfall	Why it is dangerous	Safer habit
Treating SpO_2 as the whole respiratory assessment	SpO_2 does not measure ventilation, work of breathing, or oxygen content	Always pair saturation with respiratory rate, mental status, effort, and gas when needed.
Calling all wheeze asthma/COPD	Pulmonary edema, PE, upper airway lesions, and anaphylaxis can wheeze	Use tempo, exam, CXR/ECG, and response to therapy.
Delaying decompression for tension pneumothorax	Obstructive shock can progress to arrest	Treat unstable tension physiology clinically.
Using oxygen alone for hypercapnic failure	Oxygen may improve saturation while CO_2 and acidosis worsen	Support ventilation and treat the cause.
Ordering D-dimer without pretest probability	Low specificity can drive unnecessary imaging	Use structured risk assessment first.
Missing extrapulmonary dyspnea	Anemia, acidosis, ACS, arrhythmia, and anxiety can mimic lung disease	Let physiology and initial tests broaden the differential.

Original summary diagram: bedside pulmonary reasoning map

Starting clue	Physiology	Likely diagnoses	First decisive test or action
Low SpO2 + clear lungs	Vascular, early parenchymal, low Hb/O2 content, or nonpulmonary	PE, early pneumonia, anemia, methemoglobinemia, ACS	ECG/CXR, CBC, risk-based PE testing, ABG/co-oximetry if saturation gap.
Low SpO2 + focal findings	Regional alveolar filling or collapse	Pneumonia, atelectasis, aspiration	CXR/ultrasound; antibiotics or airway clearance based on cause.
Low SpO2 + diffuse crackles	Diffuse alveolar/interstitial fluid	Pulmonary edema, ARDS, diffuse pneumonia	CXR/ultrasound, BNP/echo context, oxygen/NIV, treat cause.
High CO2 + wheeze	Ventilation failure from obstruction	COPD/asthma exacerbation	Bronchodilators/steroids, NIV when appropriate, monitor gas.
High CO2 + weak mechanics	Respiratory pump failure	Myasthenia, Guillain-Barre, ALS, opioid/sedative effect	NIF/vital capacity, medication review, early ventilatory plan.
Pleuritic pain + unilateral absent sounds	Pleural air	Pneumothorax	CXR if stable; immediate decompression if unstable.

Practice questions

Question 1. A 68-year-old with COPD has worsening somnolence, pH 7.24, PaCO2 72 mmHg, and SpO2 92% on supplemental oxygen. What is the key physiologic problem?

Answer. Ventilatory failure with respiratory acidosis. The saturation is not the main reassurance; the patient needs ventilatory support and treatment of the exacerbation trigger.

Question 2. A 31-year-old has sudden pleuritic chest pain, tachycardia, clear lungs, and a normal chest radiograph after a long flight. What diagnosis should remain high on the list?

Answer. Pulmonary embolism. A normal lung exam and CXR do not exclude PE because the problem is vascular obstruction and dead-space physiology.

Question 3. A patient with severe asthma becomes less wheezy but more fatigued and confused. Why is this concerning?

Answer. Less wheeze can mean less airflow, not improvement. Fatigue and confusion suggest impending ventilatory failure.

Question 4. Spirometry shows low FEV1/FVC, high TLC, and low DLCO. Which pattern does this suggest?

Answer. Obstructive disease with impaired gas transfer, classically emphysema-predominant COPD.

Question 5. A hypoxemic patient has minimal improvement despite high FiO2. Which mechanism should be suspected?

Answer. Large shunt physiology, such as severe atelectasis or ARDS, where perfused blood bypasses ventilated alveoli.

Coverage checklist for the next pulmonary chapters

The uploaded card file strongly signals that later pulmonary chapters should cover asthma, COPD/chronic bronchitis/emphysema, pneumonia including atypical and opportunistic infections, pulmonary embolism, pleural effusion, pneumothorax, interstitial and occupational lung disease, pulmonary hypertension, sleep-disordered breathing, lung cancer/paraneoplastic syndromes, and critical care syndromes such as ARDS. Those topics are intentionally previewed here but should receive their own disease-specific chapters so that diagnosis and management can be developed in full.

Selected references used for medical accuracy checks

- Merck Manual Professional Edition. Overview of respiratory failure; acute hypoxemic respiratory failure/ARDS. Accessed April 2026.
- Mirabile VS, et al. Respiratory Failure in Adults. StatPearls/NCBI Bookshelf. Updated 2023.
- Lagina M, et al. Diagnosis and Management of Acute Respiratory Failure. PMC, 2024.
- Johnson JD, Theurer WM. A Stepwise Approach to the Interpretation of Pulmonary Function Tests. American Family Physician. 2014.
- Ponce MC, Sankari A, Sharma S. Pulmonary Function Tests. StatPearls/NCBI Bookshelf. Updated 2023.